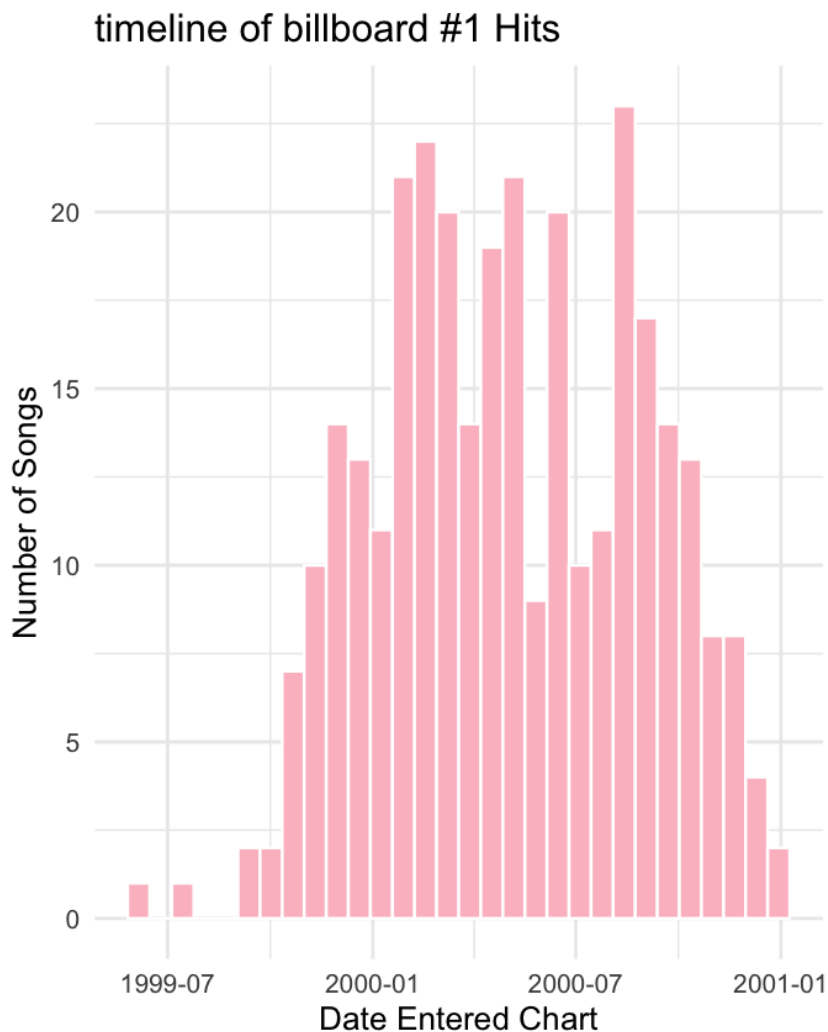


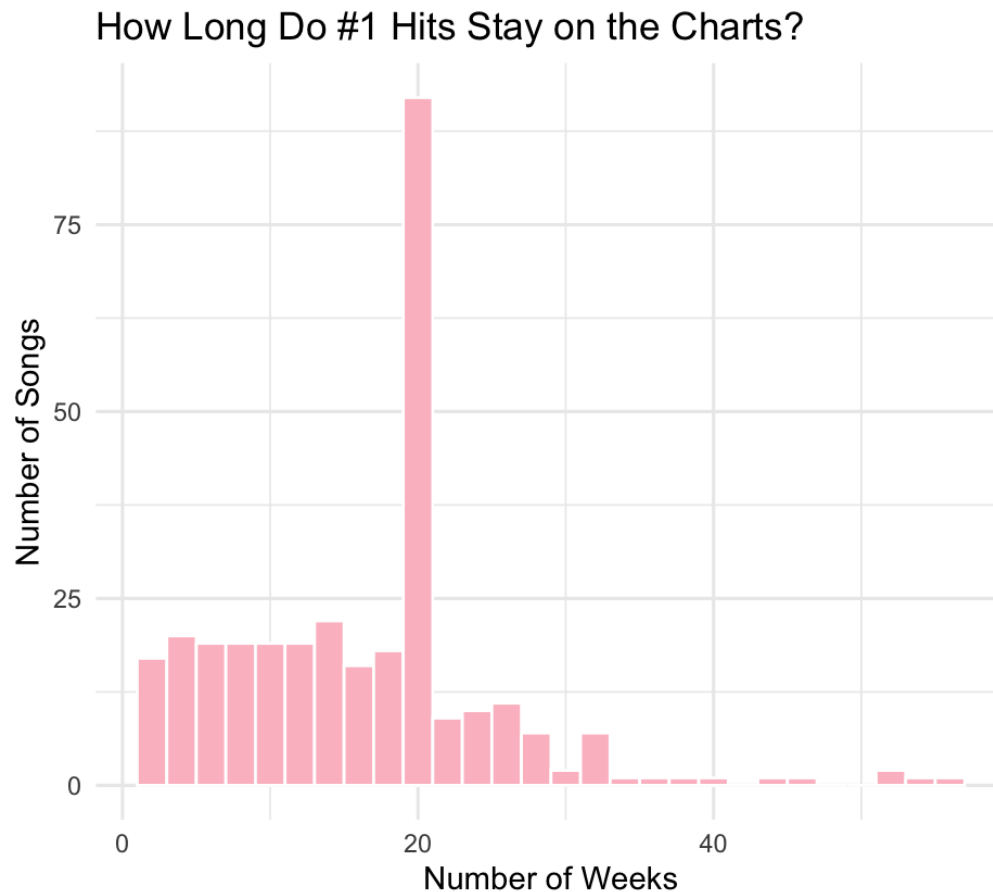
Figure #1



To make this graph I used the packages `readxl` and `tidiverse` libraries to open and use the excel based billboard dataset. Using the `read_excel ()` this allows R to read and upload the data into the R environment named as “billboard” used in future commands. I started exploring the data with the commands `glimpse(billboard)` and `summary(billboard)` to label variable types and data structure. And also check if there are any missing values using the command `colSums9(is.na(billboard))` that checks all the columns.

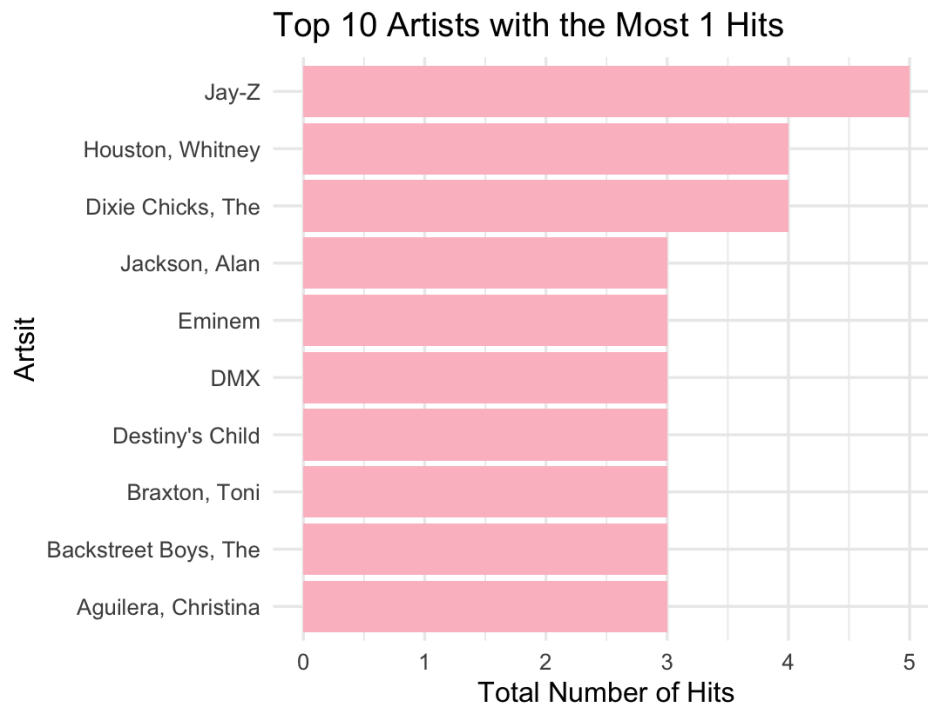
The graph is a visual representation to show the count of songs reaching number one based on dates/ year. The time frame reported is from June 1999 to December 2000 where the genre of pop was the most trending. There is a spike of hits starting in the early 2000 and peak over the course of the year. The gap in the year during 1999 shows the beginning of a genre with a few single top hits. After this, the bars show a consistent height of at least a minimum of over 13 number of hit songs.

Figure #2



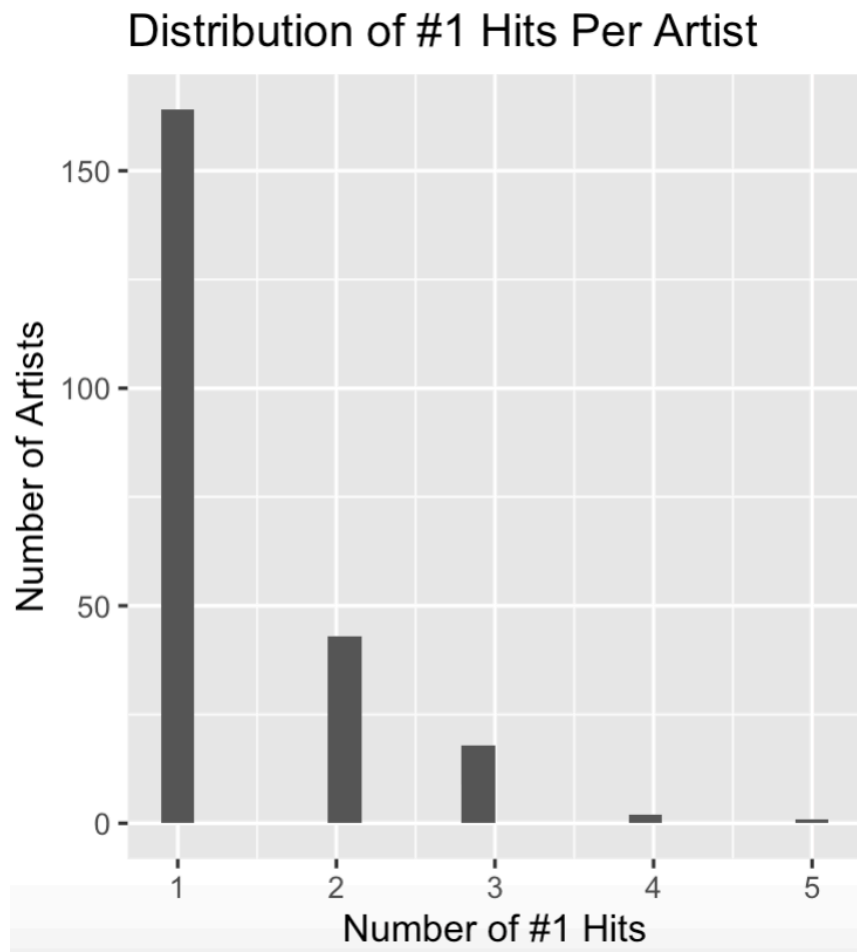
This graph is a visual representation of how many weeks a track maintains its status on the Billboard charts before dropping down. As we can see, 20 weeks identifies as a phenomenon, where songs falling below rank 50 after 20 weeks. The distribution shape shows a right skewed distribution meaning that most songs in the data have a short spotlight while only very few songs remain consistently above the average of the 20 weeks. I used the **count()** command to find the top 10 artists with the most number one hits. And calculated the average number of weeks a song spends in the number one spot to create a baseline for measuring their success. By looking at this throughout the weeks we could quantify the standard deviation and volatility of the billboards charts during the year 2000.

Figure #3

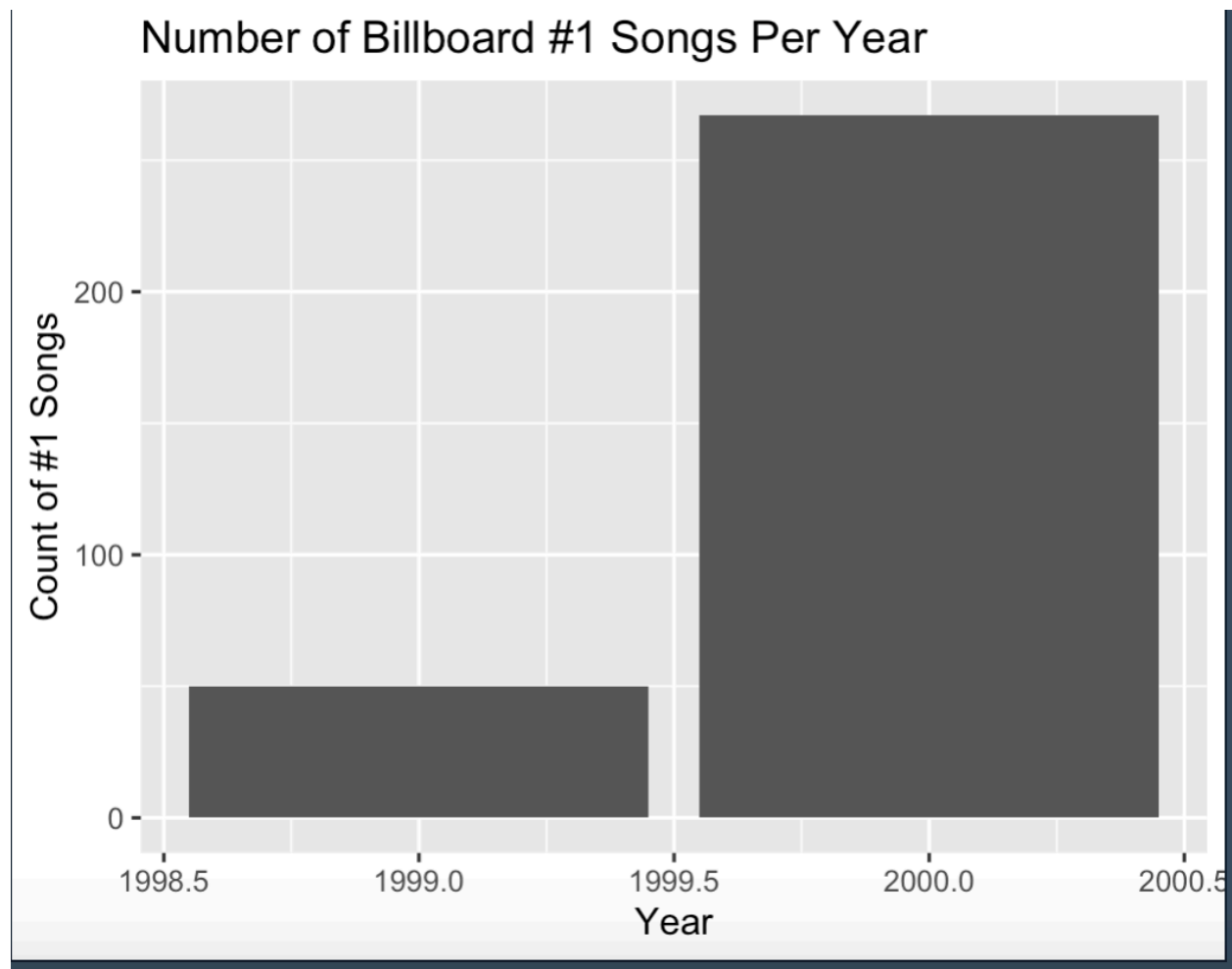


In this visualization we can see the top artists in the billboard charts, the horizontal chart specifically allows for visible artists names and easy to compare numbers, because its in a ranking stle. Jay z is the top artists ranked, with the most #1 hits among all the top ten artist. The artist that follow closely behind him include Whitney Houston, and The Dixie Chicks, that reprenstse there strong and repeated commercial and marketing usccce. Other artist such as Alan Jackson, Eminem, DMX, Destiny's Child and etc, show multiple hits but less than that top three. The contrast between these bars in the chart clearly demonstrate which artists dominate. The genres or background of these artists include mainly hip hop, country and R&B, and was not dominated by one single indysrty. The distribution of the chart is almost evenly spread out which means the media prefers this specific gorup of artists to be reliable for good music.

Figure #4



This graph shows how many artist achieved more than one single # hit within the data set time frame of mostly being in the 2000s. Most of these artists were able to achieve at least one hit, which is the tallest bar on the graph. And a sharp decline by artists with more than 2 hits and so on. The bar indicates very few artists were able to collect multiple hits. The distribution of this bar chart is mostly right skewed which means that again most categorize at having an average one hit song. This also shows that staying at the top of the billboard charts is hard to sustain, and it has it favorite artists by the public.



This visualization is about the total number of songs hitting that number one position on the billboard. This specific visual is able to show the two years of 1999 the beginning of an era and the 2000 when music, pop genre and industry was gaining momentum. The difference between these two years is significantly different, it shows how well songs and music was doing. In 1999 the number of hit songs was low, in comparison to the year 2000 where there is a high number of hit songs. This increase can be a reflection from more artists, more daily releases, and stronger commercial and media activity during the 2000's. This comparison is also able to highlight the waves of music dynamics.